

Homework 7

Domain & Range Practice, and an Introduction to Limits

Name: _____

Instructions

Use a **pen** and **show every step**. Part 1 is more practice with the domain-and-range method from last time — the goal is to make the procedure automatic. Part 2 introduces **limits** from three angles. Write questions in the margin in a different color. You may use AI as a **teacher**, but try each problem yourself first.

Some questions in part 2 may be unfamiliar to you. You should ask AI to teach you those concepts.

Part 1 — Domain and Range (more practice)

Refer to the reading from homework 6 on how to write a detailed response. You should imitate that structure for the following problems.

Method reminder. For the **domain**, avoid the two dangers: no dividing by zero, and no square roots of negatives. For the **range**, set $y = f(x)$ and **solve for** x ; the range is the set of y -values that give a valid x .

The first problem will begin on the next page.

1. $f(x) = \sqrt{x-4}$. Find the domain and range.

2. $g(x) = \frac{3x}{x+1}$. Find the domain and range.

3. $h(x) = x^2 - 5$. Find the domain and range.

Part 2 — Introduction to Limits

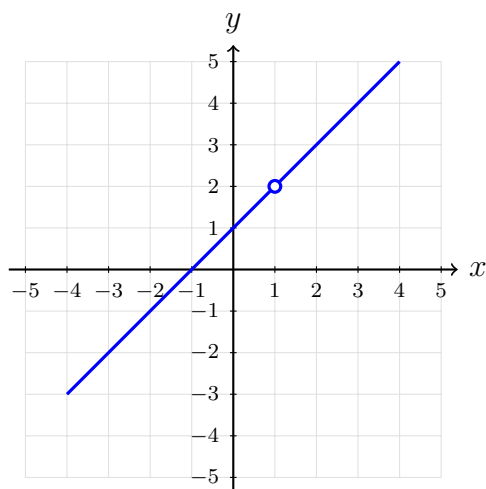
A **limit** asks: as x gets close to a number a , what value is $f(x)$ heading toward? We write

$$\lim_{x \rightarrow a} f(x).$$

The limit is about where f is *headed* as x approaches a — not its value *at* a , which may be different, or may not even exist. Below you will find a limit three ways: from a **graph**, from a **table**, and with **algebra**.

Angle 1 — Read the limit from a graph

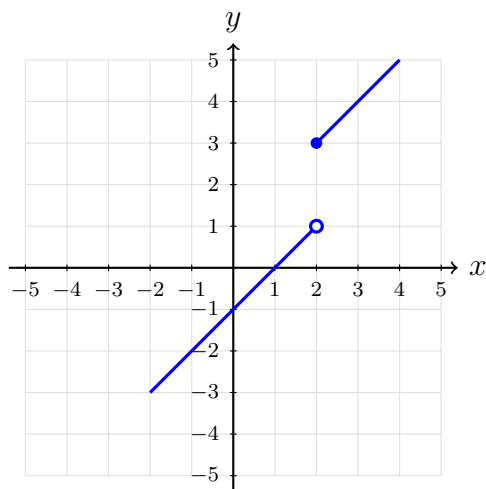
4. Use the graph of f below.



(a) $\lim_{x \rightarrow 1} f(x) =$ _____

- (b) Is $f(1)$ defined? How can you tell from the graph?

5. Use the graph of g below.



(a) Approaching $x = 2$ from the **left**, $g(x) \rightarrow$ _____

(b) From the **right**, $g(x) \rightarrow$ _____

(c) Does $\lim_{x \rightarrow 2} g(x)$ exist? Explain.

(d) $g(2) =$ _____

Angle 2 — Read the limit from a table

6. The table shows values of $f(x) = \frac{x^2 - 9}{x - 3}$ near $x = 3$. (Note: $f(3)$ is $\frac{0}{0}$, undefined.) What does $\lim_{x \rightarrow 3} f(x)$ appear to be?

x	2.9	2.99	2.999	3.001	3.01	3.1
$f(x)$	5.9	5.99	5.999	6.001	6.01	6.1

$\lim_{x \rightarrow 3} f(x) =$ _____

7. The table shows values of $h(x)$ near $x = 1$.

x	0.9	0.99	0.999	1.001	1.01	1.1
$h(x)$	1.9	1.99	1.999	4.001	4.01	4.1

(a) Approaching from the left, $h(x) \rightarrow$ _____;

(b) From the right, $h(x) \rightarrow$ _____.

(c) Does $\lim_{x \rightarrow 1} h(x)$ exist? Explain.

Angle 3 — Find the limit with algebra

First try substituting the number for x . If you get an ordinary number, that is the limit. If you get $\frac{0}{0}$, factor the top and bottom, cancel, and then substitute.

8. Find each limit. Show your steps.

(a) $\lim_{x \rightarrow 4} (3x - 5)$

(b) $\lim_{x \rightarrow 2} (x^2 + x - 1)$

(c) $\lim_{x \rightarrow 1} \frac{x^2 - 1}{x - 1}$

(d) $\lim_{x \rightarrow 3} \frac{x^2 - 9}{x - 3}$ (compare with your answer to Q6!)

(e) $\lim_{x \rightarrow 2} \frac{x^2 - 4}{x^2 - x - 2}$

Notebook Artifact

N1. If you used the internet or AI, describe how you used it.
